

OAKLAND TROY, MI, USA

WMO: 720113

Lat: 42.543N

Lon: 83.178W

Elev: 218

StdP: 98.73

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 54829

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-12.6	-21.4	0.6	-13.7	-18.7	0.7	-11.6	10.5	-2.3	9.2	-1.9	2.9	210	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	32.9	21.6	31.7	21.1	30.2	20.5	23.9	29.2	23.0	28.5	22.2	27.4	4.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.5	26.4	21.3	16.4	25.7	20.6	15.7	25.2	72.4	29.2	69.2	28.3	66.2	27.6	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
8.9	7.9	7.1	-18.4	35.3	3.6	1.3	-21.0	36.2	-23.1	37.0	-25.1	37.7	-27.7	38.6		
			-19.0	25.4	3.4	0.8	-21.5	25.9	-23.5	26.4	-25.4	26.8	-27.9	27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-2.9	-2.2	3.1	9.5	16.3	21.4	23.9	23.0	19.1	12.3	5.4	0.2
	DBStd	10.60	5.79	5.55	5.87	4.88	4.92	3.57	3.07	2.89	4.06	4.71	4.90	4.68
	HDD10.0	1528	399	343	229	67	6	0	0	0	0	28	153	304
	HDD18.3	3272	657	575	474	267	100	12	1	3	39	196	388	561
	CDD10.0	1829	1	1	14	52	201	342	431	403	272	99	14	1
	CDD18.3	533	0	0	1	2	37	103	173	148	61	8	0	0
	CDH23.3	4641	0	0	9	33	368	904	1614	1192	455	64	1	0
	CDH26.7	1552	0	0	2	4	113	313	594	374	141	11	0	0
Wind	WSAvg	3.0	3.5	3.4	3.4	3.6	2.9	2.8	2.5	2.4	2.4	3.0	3.3	3.3
Precipitation	PrecAvg	846	54	46	56	75	97	84	88	78	79	68	64	56
	PrecMax	1101	97	89	100	157	220	131	169	143	179	186	111	96
	PrecMin	598	14	16	20	15	43	34	19	19	17	7	19	23
	PrecStd	120	24	22	25	31	40	30	39	29	42	35	27	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	16.0	23.9	27.3	32.2	34.1	35.3	33.8	33.0	28.1	21.5	14.8
		MCWB	12.4	11.7	16.2	15.9	20.2	21.6	23.0	22.0	22.4	20.7	13.7	12.9
	2%	DB	10.6	11.3	18.0	23.5	29.7	32.4	32.9	32.0	30.6	25.1	17.5	11.8
		MCWB	8.6	9.0	12.2	13.9	18.8	21.1	21.8	21.1	21.3	18.0	12.0	10.0
	5%	DB	7.4	7.5	15.2	21.1	27.5	30.2	31.8	30.3	27.8	22.4	15.1	8.9
		MCWB	5.5	4.8	10.6	12.5	18.1	20.3	21.3	20.7	19.6	16.5	10.7	6.8
	10%	DB	4.6	4.9	12.3	17.9	25.0	28.0	30.1	28.8	25.9	20.1	12.7	6.5
		MCWB	3.0	2.4	8.1	11.1	16.8	19.0	20.7	20.1	18.8	15.2	9.1	4.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	12.7	16.6	17.6	22.2	23.9	25.3	24.7	24.0	21.6	15.7	13.4
		MCDB	13.7	14.2	24.1	24.0	28.1	30.4	31.5	30.4	29.9	26.2	18.5	14.7
	2%	WB	8.8	8.8	14.0	15.6	20.9	22.8	24.1	23.5	22.8	19.8	13.2	10.2
		MCDB	9.7	10.9	17.3	20.7	26.8	29.1	29.3	28.4	28.0	23.9	16.6	11.3
	5%	WB	5.4	4.7	11.4	13.9	19.7	21.9	23.1	22.5	21.6	17.6	11.3	7.2
		MCDB	7.3	7.0	14.3	19.4	25.5	27.7	28.6	27.3	25.8	21.0	14.5	8.6
	10%	WB	3.1	2.6	8.0	12.0	18.2	20.8	22.3	21.7	20.4	15.7	9.5	4.5
		MCDB	4.5	4.7	11.5	17.4	23.7	25.8	27.4	26.4	24.2	19.0	12.7	6.4
Mean Daily Temperature Range	5% DB	MDBR	6.6	7.1	8.4	10.3	10.7	10.2	10.0	9.4	9.5	8.7	7.4	5.4
		MCDBR	9.1	10.3	12.9	14.7	13.0	12.0	11.3	10.8	11.9	11.9	11.3	8.5
		MCWBR	8.1	8.5	8.6	8.0	5.9	5.0	4.2	3.9	4.6	6.3	7.7	7.6
	5% WB	MCDBR	9.1	10.1	12.3	13.2	11.1	10.5	9.7	9.0	10.0	10.5	9.9	8.1
		MCWBR	8.5	8.9	9.1	8.9	5.9	5.6	4.4	4.1	4.9	6.7	7.8	7.9
Clear-Sky Solar Irradiance	taub	0.299	0.333	0.349	0.394	0.423	0.443	0.433	0.426	0.387	0.358	0.319	0.306	
	taud	2.395	2.280	2.362	2.254	2.223	2.203	2.253	2.263	2.354	2.419	2.493	2.418	
	Ebn at Noon	842	859	893	873	853	835	838	833	842	825	815	797	
	Edh at Noon	79	104	109	131	139	142	134	128	109	90	72	71	
All-Sky Solar Radiation	RadAvg	1.49	2.37	3.52	4.61	5.38	6.08	6.09	5.26	4.25	2.75	1.70	1.21	
	RadStd	0.15	0.28	0.36	0.41	0.47	0.38	0.30	0.21	0.38	0.24	0.22	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page